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INFORMATION PROCESSING SYSTEM, ECU, AND INFORMATION PROCESSING METHOD

Abstract

In an information processing system including: a plurality of ECUs mounted on a vehicle equipped with a battery that is chargeable with an external power supply; and a communication terminal mounted on the vehicle, when the communication terminal is in a sleep state, a predetermined ECU among the plurality of ECUs activates the communication terminal, triggered by detection of start of charging of the battery with an external power supply. The communication terminal activated by the predetermined ECU acquires alarm information that is information regarding presence or absence of an alarm in a device mounted on the vehicle, from the plurality of ECUs. The communication terminal transmits the alarm information acquired from the plurality of ECUs to a predetermined server.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. application Ser. No. 17/532,419, filed Nov. 22, 2021. This application claims priority to Japanese Patent Application No. 2021-010192 filed on Jan. 26, 2021, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to an information processing system, an electronic control unit (ECU), and an information processing method.

2. Description of Related Art

[0003] An in-vehicle communication device for transmitting abnormality information to an information center when an abnormality occurs in a device mounted on a vehicle is known (see, for example, Japanese Unexamined Patent Application Publication No. 2014-234100 (JP 2014-234100 A)).

SUMMARY

[0004] An object of the present disclosure is to provide a technique capable of transmitting alarm information to a predetermined server even when the battery is charged with an external power supply.

[0005] The present disclosure can be regarded as an information processing system including: a plurality of electronic control units (ECUs) mounted on a vehicle equipped with a battery that is chargeable with an external power supply and configured to control an in-vehicle device including a battery; and a communication terminal mounted on the vehicle, the communication terminal acquiring alarm information that is information regarding presence or absence of an alarm in the in-vehicle device from the plurality of ECUs and transmitting the acquired alarm information to a predetermined server when the communication terminal is in an activated state, and the communication terminal stopping acquisition of the alarm information and transmission of the alarm information to the predetermined server when the communication terminal is in a sleep state. In the information processing system in that case, for example, when the communication terminal is in the sleep state, a predetermined ECU among the plurality of ECUs may execute detecting start of charging of the battery with an external power supply, and shifting the communication terminal from the sleep state to an activated state, triggered by detection of the start of charging of the battery with an external power supply.

[0006] The present disclosure can also be regarded as an ECU mounted on a vehicle equipped with a battery that is chargeable with an external power supply. In that case, the ECU may include a control unit that executes, for example, when a communication terminal mounted on the vehicle is in a sleep state, detecting start of charging of the battery with an external power supply, and shifting

the communication terminal from the sleep state to an activated state, triggered by detection of the start of charging of the battery with an external power supply.

[0007] The present disclosure can also be regarded as an information processing method. In the information processing method in that case, for example, an ECU mounted on a vehicle equipped with a battery that is chargeable with an external power supply may execute, when a communication terminal mounted on the vehicle is in a sleep state, a step of detecting start of charging of the battery with an external power supply, and a step of shifting the communication terminal from the sleep state to an activated state, triggered by detection of the start of charging of the battery with an external power supply.

[0008] The present disclosure can also be regarded as a program for causing the ECU to execute the above information processing method, or a non-transitory storage medium that stores the program.

[0009] According to the present disclosure, it is possible to provide a technique capable of transmitting alarm information to a predetermined server even when a battery is charged with an external power supply.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0011] FIG. 1 is a diagram showing an outline of an in-vehicle system;

[0012] FIG. 2 is a diagram showing examples of electronic control units (ECUs) mounted on a vehicle;

[0013] FIG. 3 is a diagram showing an example of a hardware configuration of components included in the in-vehicle system;

[0014] FIG. 4 is a block diagram showing an example of a functional configuration of a charging ECU;

[0015] FIG. 5 is a block diagram showing an example of a functional configuration of a communication terminal;

[0016] FIG. 6 is a sequence diagram showing an example of processes performed in the in-vehicle system when the battery is charged with an external power supply in the embodiment;

[0017] FIG. 7 is a flowchart showing a process routine performed by the charging ECU when charging of the battery with an external power supply is started;

[0018] FIG. 8 is a sequence diagram showing an example of processes performed in the in-vehicle system when the battery is charged with an external power supply in a first modification; and

[0019] FIG. 9 is a sequence diagram showing an example of processes performed in the in-vehicle system when the battery is charged with an external power supply in a second modification.

DETAILED DESCRIPTION OF EMBODIMENTS

[0020] In recent years, with the progress of vehicle communication technology such as Vehicle-to-Everything (V2X), development of a vehicle equipped with a device (communication terminal) capable of communicating with an external device has been promoted. In such a vehicle, the communication terminal can collect information (alarm information) regarding presence or absence of an alarm of devices mounted on the vehicle (hereinafter, may be referred to as “in-vehicle devices”) and transmit the collected alarm information to an external server and the like.

[0021] Here, in new energy vehicles such as electric vehicles (EVs) and plug-in hybrid vehicles (PHVs), the alarm information may be required by laws, regulations, and the like to be transmitted periodically to a server (predetermined server) operated by a public institution and the like. In

particular, for vehicles equipped with a battery that can be charged with an external power supply, the alarm information may be required to be transmitted periodically to the predetermined server when the battery is being charged with an external power supply as well as when the vehicle is in operation.

[0022] The battery is charged with an external power supply when the vehicle is not in operation. Therefore, there is a possibility that the communication terminal is in the sleep state when the battery is charged with the external power supply. The communication terminal can acquire the alarm information and transmit the alarm information to a predetermined server when the communication terminal is in the activated state. However, when the communication terminal is in the sleep state, the communication terminal cannot acquire the alarm information and transmit the alarm information to the predetermined server.

[0023] Therefore, in order to comply with the above-mentioned laws, regulations, and the like, a technique is desired that is capable of acquiring alarm information and transmitting the alarm information to a predetermined server when the battery is charged with an external power supply.

[0024] Thus, in the information processing system according to the present disclosure, when the communication terminal is in the sleep state, the communication terminal is shifted from the sleep state to the activated state, triggered by a fact that a predetermined ECU, among a plurality of ECUs mounted on a vehicle, detects the start of charging of the battery with an external power supply. This enables the communication terminal to acquire the alarm information through the plurality of ECUs and transmit the acquired alarm information to the predetermined server.

[0025] With the information processing system according to the present disclosure, it is possible to transmit the alarm information to the predetermined server even when the battery is charged with an external power supply.

[0026] Here, the predetermined ECU according to the present disclosure may be, for example, an ECU for controlling charging of the battery (first ECU), among the plurality of ECUs mounted on the vehicle. When the vehicle is not in operation, the plurality of ECUs mounted on the vehicle may be in the sleep state as is the communication terminal. However, when the charging cable is connected to the charging port of the vehicle at the start of charging of the battery with an external power supply, a detection signal of a sensor for detecting the connection of the charging cable to the charging port or a control pilot (CPLT) signal from the external power supply and the like is input to the first ECU, which activates the first ECU. In addition, when timer charging (charging of the battery at a preset time such as nighttime) is performed with the charging connector connected to the charging port of the vehicle in advance, a signal from the timer is input to the first ECU at a timing that the charging start time arrives, which activates the first ECU. Thereby, the first ECU can detect the start of charging of the battery with an external power supply. Thus, by using such a first ECU, it is possible to activate the communication terminal that was in the sleep state. As a result, the alarm information can be transmitted to the predetermined server even when the battery is charged with an external power supply.

[0027] Further, the predetermined ECU according to the present disclosure may be a second ECU that is activated by the above first ECU when the charging of the battery with an external power supply is started. The “second ECU” here is, for example, an ECU for managing the state of the battery (state of charge (SOC), temperature, and the like). When the battery is charged with an external power supply, it is necessary to properly manage the state of the battery during charging. Therefore, when the charging of the battery with an external power supply is started, the second ECU as described above is activated by the first ECU. Thereby, the second ECU can detect the start of charging of the battery with an external power supply. Thus, by using such a second ECU, it is possible to activate the communication terminal that was in the sleep state.

[0028] Further, the predetermined ECU according to the present disclosure may be a third ECU that is directly connected to the battery via wiring for detecting the voltage of the battery. This assumes a vehicle (for example, PHV) driven by a hybrid system that includes a motor and an

internal combustion engine. In such a vehicle, the ECU for controlling the hybrid system (third ECU) and the battery may be directly connected by wiring. In such a case, when the charging of the battery with an external power supply is started, the voltage input to the third ECU via the wiring changes. Thereby, the third ECU shifts from the sleep state to the activated state, triggered by the change in the voltage input via the wiring. At that time, the third ECU can detect the start of charging of the battery based on the change in the voltage input via the wiring. Thus, by using such a third ECU, it is possible to activate the communication terminal that was in the sleep state.

[0029] Here, when the predetermined ECU according to the present disclosure activates the communication terminal, the predetermined ECU may transmit to the communication terminal a first signal indicating that the charging of the battery with an external power supply has been started. Thereby, the communication terminal can start the acquisition of the alarm information and the transmission of the acquired alarm information to the predetermined server, triggered by the reception of the first signal.

[0030] When the charging of the battery with an external power supply is ended, the predetermined ECU according to the present disclosure may transmit to the communication terminal a second signal indicating that the charging of the battery with an external power supply has been ended. Thereby, the communication terminal can end the acquisition of the alarm information and the transmission of the acquired alarm information to the predetermined server, triggered by the reception of the second signal.

EMBODIMENTS

[0031] Hereinafter, an embodiment of the present disclosure will be described with reference to the drawings. The configuration of the following embodiment is illustrative, and the present disclosure is not limited to the configuration of the embodiment.

Overall Configuration of System

[0032] FIG. 1 is a diagram showing a schematic configuration of an in-vehicle system to which the present disclosure is applied. The in-vehicle system in the present embodiment includes a plurality of ECUs **100** mounted on a vehicle **10** and a communication terminal **200** mounted on the vehicle **10**.

[0033] The vehicle **10** in this example is a PHV equipped with a hybrid system including a battery that can be charged with an external power supply. The hybrid system includes a battery that can be charged with an external power supply, a motor generator that uses electric power of the battery to drive the drive wheels of the vehicle **10** or uses kinetic energy of the drive wheels to generate electricity, an internal combustion engine that cooperates with the motor generator to cause the vehicle **10** to travel (hereinafter, may be referred to as “engine”), an inverter for converting the electric power supplied from the battery to the motor generator from direct current to alternating current, or converting the electric power supplied from the motor generator to the battery from alternating current to direct current, and the like. The vehicle **10** is not limited to the PHV, and may be an EV.

[0034] The plurality of ECUs **100** represents electronic control units for controlling various in-vehicle devices such as the above-mentioned battery, motor generator, engine, and hybrid system. As shown in FIG. 2, the plurality of ECUs **100** in this example includes an ECU (motor ECU) **101** for controlling a motor generator, an ECU (engine ECU) **102** for controlling an engine, an ECU (battery ECU) **103** for managing the state of the battery (for example, SOC or temperature), an ECU (charging ECU) **104** for controlling charging of the battery, an ECU (hybrid ECU) **105** for integrally controlling the overall hybrid system, and the like. The plurality of ECUs **100** is connected to each other by a first bus CB**100** of controller area network (CAN) standard. Among the plurality of ECUs **100**, the ECUs related to the battery (battery ECU **103**, charging ECU **104**, hybrid ECU **105**, and the like) are also connected to each other by a second bus CB**200** of the CAN standard. Of the ECUs **103** to **105** related to the battery, the hybrid ECU **105** is directly connected to the battery via a wiring WR**100**, and can directly monitor the voltage of the battery.

[0035] Each of the plurality of ECUs **100** has a function of transmitting alarm information to the communication terminal **200** at a predetermined cycle when the vehicle **10** is in operation (when the ignition switch or the power switch is in ON state). Among the plurality of ECUs **100**, the ECUs **103** to **105** related to the battery have a function of transmitting the alarm information to the communication terminal **200** at a predetermined cycle during a period in which the battery is being charged with an external power supply, even when the vehicle **10** is not in operation (when the ignition switch or the power switch is in OFF state). Along with this, among the ECUs **103** to **105** related to the battery, the charging ECU **104** also has a function of activating the communication terminal **200** when the charging of the battery with an external power supply is started. Such a charging ECU **104** corresponds to the “predetermined ECU (first ECU)” according to the present disclosure. In the present embodiment, the ECUs **103** to **105** related to the battery correspond to the “plurality of ECUs” according to the present disclosure.

[0036] The communication terminal **200** is connected to the first bus CB**100** and the second bus CB**200**. The communication terminal **200** has a function of receiving the alarm information from each ECU **100** via the first bus CB**100** or the second bus CB**200**. The communication terminal **200** also has a function of aggregating the received alarm information and transmitting the aggregation result (hereinafter, may be referred to as “alarm list”) to the center server **300**. These functions are enabled when the communication terminal **200** is in the activated state, and disabled when the communication terminal **200** is in the sleep state. The center server **300** is a server operated by a public institution. The “public institution” here is, for example, an institution that requires new energy vehicles to periodically provide the alarm list. In this example, it is assumed that the alarm list is required to be periodically provided when the battery is being charged with an external power supply as well as when the vehicle **10** is in operation.

[0037] In the in-vehicle system configured as described above, when the vehicle **10** is in operation, each of the plurality of ECUs **100** transmits the alarm information to the communication terminal **200** at a predetermined cycle. At that time, the alarm information is transmitted using the first bus CB**100**. Each time the communication terminal **200** receives the alarm information from the plurality of ECUs **100**, the communication terminal **200** aggregates the alarm information to generate an alarm list and transmits the generated alarm list to the center server **300**.

[0038] Further, when the battery is charged with an external power supply, only the ECUs **103** to **105** related to the battery, among the plurality of ECUs **100**, transmit the alarm information to the communication terminal **200** at a predetermined cycle. At that time, the alarm information is transmitted using the second bus CB**200**. Each time the communication terminal **200** receives the alarm information from the ECUs **103** to **105** related to the battery, the communication terminal **200** aggregates the alarm information to generate an alarm list and transmits the generated alarm list to the center server **300**.

[0039] The battery is charged with an external power supply when the vehicle **10** is not in operation. When the vehicle **10** is not in operation, the communication terminal **200** is in the sleep state unless a special event (for example, activation of the security system) occurs. Therefore, when charging of the battery with an external power supply is started, it is necessary to shift the communication terminal **200** from the sleep state to the activated state. In contrast, in this example, the charging ECU **104** activates the communication terminal **200** when charging of the battery with an external power supply is started. As a result, even when the battery is charged with an external power supply, the communication terminal **200** can receive the alarm information and transmit the alarm list to the center server **300**.

Hardware Configuration of in-Vehicle System

[0040] FIG. **3** is a diagram showing an example of a hardware configuration of the ECU **100** and the communication terminal **200** included in the in-vehicle system. Although only the charging ECU **104**, among the plurality of ECUs **100**, is shown in FIG. **3**, the hardware configurations of the other ECUs **100** are the same.

[0041] The charging ECU **104** is an electronic control unit for controlling the in-vehicle device. The charging ECU **104** shifts from the activated state to the sleep state when the ignition switch or the power switch is switched from ON to OFF. Further, the charging ECU **104** shifts from the sleep state to the activated state when the ignition switch or the power switch is switched from OFF to ON and when various trigger signals are received or input. When the charging ECU **104** is in the activated state, the charging ECU **104** performs various processes for controlling the in-vehicle device and also performs various processes for transmitting the alarm information to the communication terminal **200** at a predetermined cycle. As shown in FIG. **3**, the charging ECU **104** includes a processor **110**, a main storage unit **120**, an auxiliary storage unit **130**, and an in-vehicle communication unit **140**. The charging ECU **104** implements a function that meets a predetermined purpose, as the processor **110** loads a program stored in the auxiliary storage unit **130** into a work area of the main storage unit **120** and executes the program.

[0042] The processor **110** is, for example, a central processing unit (CPU) or a digital signal processor (DSP). The processor **110** controls the ECU **100** and performs various information processing calculations.

[0043] The main storage unit **120** includes, for example, a random access memory (RAM), a read-only memory (ROM), and the like. As described above, in the main storage unit **120**, the work area for the processor to execute the program is set.

[0044] The auxiliary storage unit **130** includes, for example, an erasable programmable ROM (EPROM), a hard disk drive (HDD), and the like. The auxiliary storage unit **130** stores various programs, various kinds of data, and various tables in a recording medium such that they are readable and writable. The programs stored in the auxiliary storage unit **130** include, in addition to an operating system (OS) and the like, a program for implementing a function of transmitting the alarm information to the communication terminal **200** at a predetermined cycle. Such a program is common to all of the plurality of ECUs **100**. In addition, the auxiliary storage unit **130** also stores a program specific to the charging ECU **104**. The program specific to the charging ECU **104** is, for example, a program for implementing a function of activating the communication terminal **200** when charging of the battery with an external power supply is started. Part or all of the information stored in the auxiliary storage unit **130** may be stored in the main storage unit **120**. Further, part of the information stored in the main storage unit **120** may be stored in the auxiliary storage unit **130**.

[0045] The in-vehicle communication unit **140** is an interface for connecting the ECU **100** to the first bus CB**100** or the second bus CB**200**. The in-vehicle communication unit **140** is connected to the first bus CB**100** or the second bus CB**200** using, for example, a CAN standard interface circuit. In the present embodiment, the in-vehicle communication unit **140** communicates with another ECU **100** and the communication terminal **200** through the first bus CB**100** or the second bus CB**200**.

[0046] The series of processes executed by the ECU **100** configured as described above can be executed by hardware or software.

[0047] Next, the communication terminal **200** is a terminal having a function of transmitting and receiving data to and from the plurality of ECUs **100** and a function of transmitting and receiving data to and from the center server **300**. The communication terminal **200** shifts from the activated state to the sleep state when the ignition switch or the power switch is switched from ON to OFF. Further, the communication terminal **200** shifts from the sleep state to the activated state when the ignition switch or the power switch is switched from OFF to ON and when various trigger signals are received or input. When the communication terminal **200** is in the activated state, the communication terminal **200** performs various processes for transmitting the alarm list to the center server **300** at a predetermined cycle. As shown in FIG. **3**, the communication terminal **200** includes a processor **210**, a main storage unit **220**, an auxiliary storage unit **230**, an in-vehicle communication unit **240**, an out-of-vehicle communication unit **250**, and the like. The communication terminal **200** implements a function that meets a predetermined purpose, as the

processor **210** loads a program stored in the auxiliary storage unit **230** into a work area of the main storage unit **220** and executes the program.

[0048] Since the processor **210**, the main storage unit **220**, and the in-vehicle communication unit **240** are the same as the processor **110**, the main storage unit **120**, and the in-vehicle communication unit **140** of the ECU **100**, the description thereof will be omitted.

[0049] The auxiliary storage unit **230** may include a removable medium in addition to the same configuration as that of the auxiliary storage unit **130** of the ECU **100**. The removable medium is, for example, a universal serial bus (USB) memory or a disc recording medium such as a compact disc (CD) or a digital versatile disc (DVD). Further, the program stored in the auxiliary storage unit **230** includes a program for implementing a function for transmitting the alarm list to the center server **300** when the communication terminal **200** is in the activated state.

[0050] The out-of-vehicle communication unit **250** is an interface for connecting the communication terminal **200** to the network. The out-of-vehicle communication unit **250** connects to the network by a mobile communication method such as long term evolution (LTE), LTE-advanced, and 5th generation (5G), or a wireless communication method such as Wi-Fi. In the present embodiment, the out-of-vehicle communication unit **250** communicates with the center server **300** through the network.

[0051] The series of processes executed by the communication terminal **200** configured as described above can be executed by hardware or software.

Functional Configuration of ECU

[0052] Here, an example of the functional configuration of the charging ECU **104** in the present embodiment will be described with reference to FIG. 4. As shown in FIG. 4, the charging ECU **104** in the present embodiment includes a charging processing unit **F1041**, a detection unit **F1042**, an activation processing unit **F1043**, and a data transmission unit **F1044** as its functional components. The charging processing unit **F1041**, the detection unit **F1042**, the activation processing unit **F1043**, and the data transmission unit **F1044** are implemented, as the processor **110** executes a program loaded on the main storage unit **120**. These functional components work when the charging ECU **104** is in the activated state and do not work when the charging ECU **104** is in the sleep state. Among the functional components of the charging ECU **104**, the combination of the detection unit **F1042** and the activation processing unit **F1043** corresponds to the “control unit” of the ECU according to the present disclosure.

[0053] The charging processing unit **F1041** performs various processes for charging the battery with an external power supply. Here, when the vehicle **10** is not in operation, the plurality of ECUs **100** including the charging ECU **104** is also in the sleep state similarly to the communication terminal **200**. In such a state, when a charging cable from the external power supply is connected to the charging port of the vehicle **10**, a detection signal of a sensor for detecting connection of the charging cable to the charging port or a trigger signal such as a CPLT signal from the external power supply is input to the charging ECU **104**. Examples of the above-mentioned sensor include a limit switch, a proximity sensor, a photoelectric sensor, and the like. When timer-charging is performed with the charging connector connected to the charging port of the vehicle **10** in advance, a trigger signal is input from the timer to the charging ECU **104** at the timing when the charging start time arrives. When the various trigger signals described above are input to the charging ECU **104**, the charging ECU **104** shifts from the sleep state to the activated state. The charging ECU **104** activated in this way starts charging of the battery from the external power supply. For example, the charging processing unit **F1041** switches a relay provided between the charging port of the vehicle **10** and the battery from OFF (a state in which the charging port and the battery are not electrically connected) to ON (a state in which the charging port and the battery are electrically connected), thereby starting charging of the battery from the external power supply. Further, when starting charging of the battery from the external power supply, the charging processing unit **F1041** transmits a trigger signal for shifting the battery ECU **103** from the sleep state to the activated state

through the second bus CB200. Thereby, when the battery is being charged with an external power supply, the charging processing unit F1041 can acquire the SOC or the temperature and the like of the battery through the battery ECU 103. As a result, the charging processing unit F1041 can control the magnitude and the like of the current flowing from the external power supply to the battery based on the information acquired from the battery ECU 103. Further, when the SOC of the battery reaches a predetermined value, the charging processing unit F1041 can end the charging of the battery with the external power supply by switching the above-mentioned relay from ON to OFF.

[0054] The detection unit F1042 detects start of charging and end of charging of the battery with an external power supply. For example, the detection unit F1042 detects the start of charging and the end of charging of the battery with an external power supply by monitoring the state of the above-mentioned relay. That is, when the relay is switched from OFF to ON, the detection unit F1042 determines that charging of the battery with an external power supply has been started. When the relay is switched from ON to OFF, the detection unit F1042 determines that charging of the battery with an external power supply has been ended. The charging processing unit F1041 may notify the detection unit F1042 of the start of charging and the end of charging of the battery with an external power supply. When the detection unit F1042 detects the start of charging and the end of charging of the battery with an external power supply, the detection result is passed to the activation processing unit F1043.

[0055] The activation processing unit F1043 transmits a trigger signal for shifting the communication terminal 200 from the sleep state to the activated state to the communication terminal 200 through the second bus CB200 when the start of charging of the battery with an external power supply is detected. Following the trigger signal described above, the activation processing unit F1043 transmits a signal (first signal) indicating that charging of the battery with an external power supply has been started, to the communication terminal 200 through the second bus CB200. Further, when the end of charging of the battery with an external power supply is detected, the activation processing unit F1043 transmits a signal (second signal) indicating that the charging of the battery with an external power supply has been ended, to the communication terminal 200 through the second bus CB200.

[0056] The data transmission unit F1044 generates the alarm information and transmits the generated alarm information to the communication terminal 200. At that time, when the vehicle 10 is in operation, the data transmission unit F1044 transmits the alarm information to the communication terminal 200 through the first bus CB100. Further, when the battery is being charged with an external power supply, the data transmission unit F1044 transmits the alarm information to the communication terminal 200 through the second bus CB200. The generation of the alarm information and the transmission of the alarm information by the data transmission unit F1044 to the communication terminal 200 are repeatedly executed at a predetermined cycle. The alarm information generated by the data transmission unit F1044 is information indicating the presence or absence of an alarm related to the in-vehicle device under the control of the charging ECU 104. Therefore, the data transmission unit F1044 determines whether an alarm is generated in the above-mentioned in-vehicle device based on detection values of various sensors attached to the in-vehicle device under the control of the charging ECU 104 at the timing of generating the alarm information. Then, the data transmission unit F1044 generates the alarm information in accordance with the above determination result.

[0057] Among the functions of the data transmission unit F1044 described above, the function of generating the alarm information and transmitting the alarm information to the communication terminal 200 when the vehicle 10 is in operation is common to all of the plurality of ECUs 100. Further, among the functions of the data transmission unit F1044 described above, a function of generating the alarm information and transmitting the alarm information to the communication terminal 200 when the battery is being charged with an external power supply is common to only

the ECUs **103** to **105** related to the battery.

[0058] Any one or part of the charging processing unit **F1041**, the detection unit **F1042**, the activation processing unit **F1043**, and the data transmission unit **F1044** may be implemented by a hardware circuit. Further, the functional configuration of the ECU **100** is not limited to the example shown in FIG. **4**, and components may be omitted, replaced, or added as appropriate.

Functional Configuration of Communication Terminal

[0059] Next, an example of the functional configuration of the communication terminal **200** in the present embodiment will be described with reference to FIG. **5**. As shown in FIG. **5**, the communication terminal **200** in the present embodiment includes a reception unit **F210**, a data processing unit **F220**, and a transmission unit **F230** as its functional components. The reception unit **F210**, the data processing unit **F220**, and the transmission unit **F230** are implemented, as the processor **210** executes a program loaded on the main storage unit **220**. The reception unit **F210**, the data processing unit **F220**, and the transmission unit **F230** work when the communication terminal **200** is in the activated state, and do not work when the communication terminal **200** is in the sleep state.

[0060] The reception unit **F210** receives the alarm information transmitted from each ECU **100** through the in-vehicle communication unit **240**. At that time, when the vehicle **10** is in operation, the reception unit **F210** receives the alarm information from all of the plurality of ECUs **100**. Further, when the battery is being charged with an external power supply, the reception unit **F210** receives the alarm information only from the ECUs **103** to **105** related to the battery, among the plurality of ECUs **100**. The alarm information is repeatedly received at a predetermined cycle during the period in which the vehicle **10** is in operation and during the period in which the battery is being charged with an external power supply. A plurality of pieces of alarm information received by the reception unit **F210** is passed from the reception unit **F210** to the data processing unit **F220**.

[0061] The data processing unit **F220** generates the alarm list based on the alarm information received by the reception unit **F210**. The alarm list is a list required to be transmitted to the center server **300** at a predetermined cycle, and includes a plurality of items for registering presence or absence of an alarm of a type stipulated by laws, regulations, and the like, as described above. Thus, the data processing unit **F220** extracts information to be registered in each item of the alarm list, from the plurality of pieces of alarm information received by the reception unit **F210**, and generates the alarm list based on the extracted information. The alarm list generated by the data processing unit **F220** is passed from the data processing unit **F220** to the transmission unit **F230**.

[0062] The transmission unit **F230** transmits the alarm list generated by the data processing unit **F220** to the center server **300** through the out-of-vehicle communication unit **250**.

[0063] Any one or part of the reception unit **F210**, the data processing unit **F220**, and the transmission unit **F230** may be implemented by a hardware circuit when the communication terminal **200** is activated by the charging ECU **104**. Further, the functional configuration of the communication terminal **200** is not limited to the example shown in FIG. **5**, and components may be omitted, replaced, or added as appropriate.

Processing Flow

[0064] Next, an example of the process sequence in the in-vehicle system of the present embodiment will be described with reference to FIG. **6**. FIG. **6** is a sequence diagram showing an example of processes performed in the in-vehicle system when the battery is charged with an external power supply. In the example shown in FIG. **6**, it is assumed that the vehicle **10** is in operation and the ECUs **103** to **105** related to the battery and the communication terminal **200** are in the sleep state.

[0065] When charging of the battery with an external power supply is started, a trigger signal from the sensor for detecting the connection of the charging connector to the charging port, the external power supply, or the timer is input to the charging ECU **104**, as described above. When receiving input of the trigger signal, the charging ECU **104** shifts from the sleep state to the activated state

(S11). Accordingly, the charging ECU **104** is brought into a state of being able to control charging of the battery with an external power supply and into a state of being able to transmit the alarm information to the communication terminal **200**.

[0066] In the charging ECU **104** that has shifted from the sleep state to the activated state, the charging processing unit **F1041** transmits a trigger signal for shifting the battery ECU **103** from the sleep state to the activated state to the battery ECU **103** through the second bus **CB200** (S12).

[0067] The battery ECU **103** that has received the trigger signal from the charging ECU **104** shifts from the sleep state to the activated state (S13). Thereby, the battery ECU **103** is brought into a state of being able to transmit information regarding the battery state (SOC, or temperature and the like) to the hybrid ECU **105** and into a state of being able to transmit the alarm information to the communication terminal **200**.

[0068] The charging processing unit **F1041** of the charging ECU **104** starts charging of the battery with an external power supply after activating the battery ECU **103** (S14). Specifically, as described above, the charging processing unit **F1041** switches the relay provided between the charging port of the vehicle **10** and the battery from OFF to ON.

[0069] When the relay is switched from OFF to ON, the detection unit **F1042** of the charging ECU **104** determines that charging of the battery with an external power supply has been started. Such a determination result is passed from the detection unit **F1042** to the activation processing unit **F1043**. The activation processing unit **F1043** transmits a trigger signal for shifting the communication terminal **200** from the sleep state to the activated state to the communication terminal **200** through the second bus **CB200** (S15).

[0070] The communication terminal **200** that has received the trigger signal from the charging ECU **104** shifts from the sleep state to the activated state (S16). Thereby, the communication terminal **200** is brought into a state of being able to communicate with the ECUs **103** to **105** related to the battery and into a state of being able to communicate with the center server **300**.

[0071] After activating the communication terminal **200**, the activation processing unit **F1043** of the charging ECU **104** transmits the first signal to the communication terminal **200** through the second bus **CB200** (S17). The first signal is a signal indicating that charging of the battery with an external power supply has been started. Thereby, the communication terminal **200** is brought into a state of being able to accept the alarm information from the ECUs **103** to **105** related to the battery and into a state of being able to generate the alarm list.

[0072] Here, as described above, the hybrid ECU **105** is directly connected to the battery via a wiring **WR100**. Therefore, when the voltage of the battery changes with the start of charging of the battery with an external power supply, the voltage input to the hybrid ECU **105** via the wiring **WR100** also changes. Thereby, the hybrid ECU **105** shifts from the sleep state to the activated state, triggered by the change in the input voltage (S18). As a result, the hybrid ECU **105** is brought into a state of being able to transmit the alarm information to the communication terminal **200**, similarly to the charging ECU **104** and the battery ECU **103**.

[0073] When each of the battery ECU **103**, the charging ECU **104**, and the hybrid ECU **105** is activated, each of the ECUs **103** to **105** transmits the alarm information to the communication terminal **200** through the second bus **CB200** (S19).

[0074] When the alarm information transmitted from each of the battery ECU **103**, the charging ECU **104**, and the hybrid ECU **105** is received by the in-vehicle communication unit **240** of the communication terminal **200**, the alarm information is passed to the data processing unit **F220** through the reception unit **F210**. The data processing unit **F220** generates the alarm list based on the above alarm information (S20).

[0075] When the alarm list is generated by the data processing unit **F220**, the transmission unit **F230** of the communication terminal **200** transmits the alarm list to the center server **300** through the out-of-vehicle communication unit **250** (S21).

[0076] The above processes related to S19 to S21 are repeatedly executed at a predetermined cycle

until the charging of the battery with an external power supply ends.

[0077] When the SOC acquired through the battery ECU **103** reaches a predetermined value, the charging processing unit **F1041** of the charging ECU **104** ends the charging of the battery with an external power supply (**S22**). Specifically, the charging processing unit **F1041** switches the above-mentioned relay from ON to OFF.

[0078] When the relay is switched from ON to OFF, the detection unit **F1042** of the charging ECU **104** determines that charging of the battery with an external power supply has been ended. Such a determination result is passed from the detection unit **F1042** to the activation processing unit **F1043**. The activation processing unit **F1043** transmits a second signal to the communication terminal **200** through the second bus **CB200** (**S23**). The second signal is a signal indicating that charging of the battery with an external power supply has been ended. Thereby, the communication terminal **200** ends the acceptance of the alarm information from the ECUs **103** to **105** related to the battery and ends the generation of the alarm list. At that time, the communication terminal **200** may shift from the activated state to the sleep state.

[0079] When the charging of the battery with an external power supply is ended, a signal similar to the second signal may be transmitted from the charging ECU **104** to the battery ECU **103** and the hybrid ECU **105**. In that case, the battery ECU **103** and the hybrid ECU **105** can end the generation of the alarm information and the transmission of the alarm information to the communication terminal **200**, triggered by the reception of the above signal. Further, the battery ECU **103** and the hybrid ECU **105** can also shift from the activated state to the sleep state, triggered by the reception of the above signal.

[0080] Next, the flow of processing performed by the charging ECU **104** when the battery is charged with an external power supply will be described with reference to FIG. 7. FIG. 7 is a flowchart showing a process routine performed by the charging ECU **104** when charging of the battery with an external power supply is started. That is, FIG. 7 is a flowchart showing a process routine performed by the charging ECU **104** in **S14** and subsequent steps in FIG. 6 described above.

[0081] In FIG. 7, when the above-mentioned relay is switched from OFF to ON by the charging processing unit **F1041** of the charging ECU **104** to start the charging of the battery with an external power supply, the detection unit **F1042** detects the start of charging of the battery with an external power supply (step **S101**).

[0082] When the start of charging of the battery with an external power supply is detected in step **S101**, the activation processing unit **F1043** transmits a trigger signal for shifting the communication terminal **200** from the sleep state to the activated state, to the communication terminal **200** through the in-vehicle communication unit **140** (step **S102**). At that time, the in-vehicle communication unit **140** transmits the trigger signal to the communication terminal **200** using the second bus **CB200**. Thereby, the communication terminal **200** shifts from the sleep state to the activated state.

[0083] When the communication terminal **200** is activated in step **S102**, the activation processing unit **F1043** transmits the first signal to the communication terminal **200** through the in-vehicle communication unit **140** and the second bus **CB200** (step **S103**). The first signal is a signal indicating that the charging of the battery with an external power supply has been started, as described above. Thereby, the communication terminal **200** is brought into a state of being able to accept the alarm information from the ECUs **103** to **105** related to the battery and into a state of being able to generate the alarm list.

[0084] When the first signal is transmitted to the communication terminal **200** in step **S103**, the data transmission unit **F1044** generates the alarm information and transmits the generated alarm information to the communication terminal **200** at a predetermined cycle (step **S104**). At that time, the data transmission unit **F1044** transmits the alarm information to the communication terminal **200** through the in-vehicle communication unit **140** and the second bus **CB200**. The same

processes are performed in the battery ECU **103** and the hybrid ECU **105**, which enables the communication terminal **200** to generate the alarm list and transmit the alarm list to the center server **300** based on the alarm information received from the ECUs **103** to **105** related to the battery.

[0085] When the process of step **S104** ends, the detection unit **F1042** determines whether the charging of the battery with an external power supply has been ended (step **S105**). At that time, when the above-mentioned relay has not been switched from ON to OFF by the charging processing unit **F1041** (the relay remains ON), the detection unit **F1042** determines that the charging of the battery with an external power supply has not been ended (negative determination in step **S105**). In that case, the process of step **S104** is executed again. In contrast, when the above-mentioned relay has been switched from ON to OFF by the charging processing unit **F1041**, the detection unit **F1042** determines that the charging of the battery with an external power supply has been ended (affirmative determination in step **S105**). In that case, the process of step **S106** is executed.

[0086] In step **S106**, the activation processing unit **F1043** transmits the second signal to the communication terminal **200** through the in-vehicle communication unit **140** and the second bus **CB200**. The second signal is a signal indicating that the charging of the battery with an external power supply has been ended, as described above. Thereby, the communication terminal **200** ends the acceptance of the alarm information from the ECUs **103** to **105** related to the battery and ends the generation of the alarm list. When the process of step **S106** ends, the execution of this process routine ends.

[0087] When the processes shown in FIGS. **6** and **7** are performed, the communication terminal **200** can be shifted from the sleep state to the activated state when the battery is charged with an external power supply. This enables the communication terminal **200** to accept the alarm information from the ECUs **103** to **105** related to the battery and generate the alarm list based on the alarm information. As a result, the alarm list can be transmitted to the center server **300** even when the battery is charged with an external power supply.

First Modification

[0088] In the above-described embodiment, an example in which the charging ECU **104** activates the communication terminal **200** when the battery is charged with an external power supply has been described. However, the battery ECU **103** may activate the communication terminal **200**.

[0089] When the charging of the battery with an external power supply is started, the SOC and the voltage of the battery change. When the charging of the battery with an external power supply is ended, the SOC and the voltage of the battery stop changing. Here, as described in the above embodiment, the battery ECU **103** is activated by the charging ECU **104** before the charging of the battery with an external power supply is started. Therefore, the battery ECU **103** can detect the start of charging and the end of charging of the battery with an external power supply based on the SOC or the voltage of the battery.

[0090] Thus, in this modification, when the battery is charged with an external power supply, the battery ECU **103** detects the start of charging and the end of charging of the battery with an external power supply to activate the communication terminal **200** and transmit the first signal and the second signal to the communication terminal **200**. In this modification, the battery ECU **103** corresponds to the “predetermined ECU (second ECU)” according to the present disclosure.

[0091] Here, an example of the process sequence in the in-vehicle system of this modification will be described with reference to FIG. **8**. FIG. **8** is a sequence diagram showing an example of processes performed in the in-vehicle system when the battery is charged with an external power supply. In FIG. **8**, the same processes as in FIG. **6** are designated by the same reference signs as in FIG. **6**. Therefore, the description of the same processes as in FIG. **6** will be omitted.

[0092] In FIG. **8**, when the charging ECU **104** starts charging of the battery with an external power supply after the battery ECU **103** is activated by the trigger signal from the charging ECU **104**

(S14), the SOC and the voltage of the battery change. At that time, the charging ECU **104** detects the change in the SOC or the voltage of the battery so as to detect that the charging of the battery with an external power supply has been started (S31).

[0093] The battery ECU **103** that has detected the start of charging of the battery with an external power supply transmits a trigger signal for shifting the communication terminal **200** from the sleep state to the activated state to the communication terminal **200** through the second bus CB200 (S32). Thereby, the communication terminal **200** is activated (S16), which enables communication between the ECUs **103** to **105** related to the battery and the communication terminal **200**.

[0094] After activating the communication terminal **200**, the battery ECU **103** transmits the first signal to the communication terminal **200** through the second bus CB200 (S33). Thereby, the communication terminal **200** is brought into a state of being able to accept the alarm information from the ECUs **103** to **105** related to the battery and into a state of being able to generate the alarm list.

[0095] During a period from when the first signal is transmitted from the battery ECU **103** to the communication terminal **200** until the charging of the battery with an external power supply is ended, the transmission of the alarm information from the ECUs **103** to **105** to the communication terminal **200**, the generation of the alarm list by the communication terminal **200**, and the transmission of the alarm list from the communication terminal **200** to the center server **300** are repeated at a predetermined cycle, as in FIG. 6. When the charging ECU **104** ends the charging of the battery with an external power supply (S22), the SOC and the voltage of the battery stop changing. At that time, the battery ECU **103** detects that the SOC or the voltage of the battery has stopped changing so as to detect that the charging of the battery with an external power supply has been ended (S34).

[0096] The battery ECU **103** that has detected the end of charging of the battery with an external power supply transmits the second signal to the communication terminal **200** through the second bus CB200 (S35).

[0097] According to the above-described modification, when the battery is charged with an external power supply, the communication terminal **200** can be activated using the battery ECU **103** that is activated by the charging ECU **104**. As a result, the same effects as those of the above-described embodiment can be obtained.

Second Modification

[0098] In the first modification, an example has been described in which when the battery is charged with an external power supply, the communication terminal **200** is activated using the battery ECU **103** instead of the charging ECU **104**. However, the communication terminal **200** may be activated using the hybrid ECU **105** instead of the charging ECU **104**.

[0099] As described in the embodiment, when the charging of the battery with an external power supply is started, the voltage input to the hybrid ECU **105** via the wiring WR100 changes, which triggers activation of the hybrid ECU **105**. Thereby, the hybrid ECU **105** can determine that the charging of the battery with an external power supply has been started on the condition that the voltage input via the wiring WR100 has changed. That is, the hybrid ECU **105** can detect the start of charging of the battery with an external power supply, similarly to the battery ECU **103** described above. Further, when the charging of the battery with an external power supply is ended, the voltage of the battery stops changing, so that the voltage input to the hybrid ECU **105** via the wiring WR100 also stops changing. Thereby, the hybrid ECU **105** can determine that the charging of the battery with an external power supply has been ended on the condition that the voltage input via the wiring WR100 has stopped changing. That is, the hybrid ECU **105** can also detect the end of charging of the battery with an external power supply, similarly to the battery ECU **103** described above.

[0100] Thus, in this modification, when the battery is charged with an external power supply, the hybrid ECU **105** detects the start of charging and the end of charging of the battery with an external

power supply to activate the communication terminal **200** and transmit the first signal and the second signal to the communication terminal **200**. In this modification, the hybrid ECU **105** corresponds to the “predetermined ECU (third ECU)” according to the present disclosure.

[0101] Here, an example of the process sequence in the in-vehicle system of this modification will be described with reference to FIG. **9**. FIG. **9** is a sequence diagram showing an example of processes performed in the in-vehicle system when the battery is charged with an external power supply. In FIG. **9**, the same processes as in FIG. **6** are designated by the same reference signs as in FIG. **6**. Therefore, the description of the same processes as in FIG. **6** will be omitted.

[0102] In FIG. **9**, when the hybrid ECU **105** is activated as the charging ECU **104** starts charging of the battery with an external power supply (**S18**), the hybrid ECU **105** determines that the charging of the battery with an external power supply has been started on the condition that the voltage input via the wiring **WR100** changes. Thereby, the hybrid ECU **105** can detect the start of charging of the battery with an external power supply (**S41**).

[0103] The hybrid ECU **105** that has detected the start of charging of the battery with an external power supply transmits a trigger signal for shifting the communication terminal **200** from the sleep state to the activated state to the communication terminal **200** through the second bus **CB200** (**S42**). Thereby, the communication terminal **200** is activated (**S16**), which enables communication between the ECUs **103** to **105** related to the battery and the communication terminal **200**.

[0104] After activating the communication terminal **200**, the hybrid ECU **105** transmits the first signal to the communication terminal **200** through the second bus **CB200** (**S43**). Thereby, the communication terminal **200** is brought into a state of being able to accept the alarm information from the ECUs **103** to **105** related to the battery and into a state of being able to generate the alarm list.

[0105] During a period from when the first signal is transmitted from the hybrid ECU **105** to the communication terminal **200** until the charging of the battery with an external power supply is ended, the transmission of the alarm information from the ECUs **103** to **105** to the communication terminal **200**, the generation of the alarm list by the communication terminal **200**, and the transmission of the alarm list from the communication terminal **200** to the center server **300** are repeated at a predetermined cycle, as in FIG. **6**. When the charging ECU **104** ends the charging of the battery with an external power supply (**S22**), the voltage of the battery stops changing, and accordingly, the voltage input to the hybrid ECU **105** via the wiring **WR100** also stops changing. Thereby, the hybrid ECU **105** can determine that the charging of the battery with an external power supply has been ended on the condition that the voltage input via the wiring **WR100** has stopped changing. That is, the hybrid ECU **105** can detect the end of charging of the battery with an external power supply (**S44**).

[0106] The hybrid ECU **105** that has detected the end of charging of the battery with an external power supply transmits the second signal to the communication terminal **200** through the second bus **CB200** (**S45**).

[0107] According to the above-described modification, when the battery is charged with an external power supply, the communication terminal **200** can be activated using the hybrid ECU **105** that is autonomously activated, triggered by the start of charging of the battery with an external power supply. As a result, the same effects as those of the above-described embodiment can be obtained. Others

[0108] The above-described embodiment and modifications are merely examples, and the present disclosure may be appropriately modified to be implemented without departing from the scope thereof. Also, the processes and the configurations described in the present disclosure can be appropriately combined to be implemented as long as no technical contradiction occurs. Moreover, the processes described as being executed by one device may be shared and executed by a plurality of devices. Alternatively, the processes described as being executed by different devices may be executed by one device. In the computer system, it is possible to flexibly change the hardware

configuration for implementing each function.

[0109] The present disclosure can also be implemented by supplying a computer with a computer program that implements the functions described in the above embodiment and modifications and causing one or more processors of the computer to read and execute the program. Such a computer program may be provided to the computer by a non-transitory computer-readable storage medium connectable to the system bus of the computer, or may be provided to the computer via a network. The non-transitory computer-readable storage medium is a recording medium that can store information such as data and programs by electrical, magnetic, optical, mechanical, or chemical action and can be read from a computer or the like. Such a non-transitory computer-readable storage medium is a disc of any type such as a magnetic disc (floppy (registered trademark) disc, HDD, and the like) or an optical disc (CD-ROM, DVD, Blu-ray disc, and the like). Further, the non-transitory computer-readable storage medium may be a medium such as a read-only memory (ROM), a random access memory (RAM), an EPROM, an EEPROM, a magnetic card, a flash memory, an optical card, or a solid state drive (SSD).

Claims

1. An information processing method for electronic control units (ECUs) mounted on a vehicle equipped with a battery that is chargeable with an external power supply, the method comprising: by a first ECU, when a communication terminal mounted on the vehicle is in a sleep state, sending, to a second ECU, a trigger signal to wake up the second ECU in a sleep state, by at least one of the first ECU and the second ECU, when the communication terminal is in the sleep state, detecting start of charging of the battery with an external power supply, and by at least one of the first ECU or the second ECU, when the communication terminal is in the sleep state, shifting the communication terminal from the sleep state to an activated state, triggered by detection of the start of charging of the battery with an external power supply, wherein the ECUs include at least the first ECU and the second ECU, the first ECU is an ECU for controlling the charging of the battery, and is a ECU that is activated when the charging of the battery with an external power supply is started, the second ECU is an ECU for managing a state of the battery, and the communication terminal is configured to acquiring, from the ECUs, alarm information that is information regarding presence or absence of an alarm in an in-vehicle device including the battery and transmitting the acquired alarm information to a predetermined server when the communication terminal is in the activated state, and stopping acquisition of the alarm information and transmission of the alarm information to the predetermined server when the communication terminal is in the sleep state.
 2. The information processing method according to claim 1, further comprising executing, by at least one of the ECUs, when the communication terminal is shifted from the sleep state to the activated state, a step of transmitting to the communication terminal a first signal indicating that the charging of the battery with an external power supply has been started.
 3. The information processing method according to claim 2, further comprising executing, by at least one of the ECUs, when the charging of the battery with an external power supply is ended, a step of transmitting to the communication terminal a second signal indicating that the charging of the battery with an external power supply has been ended.
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